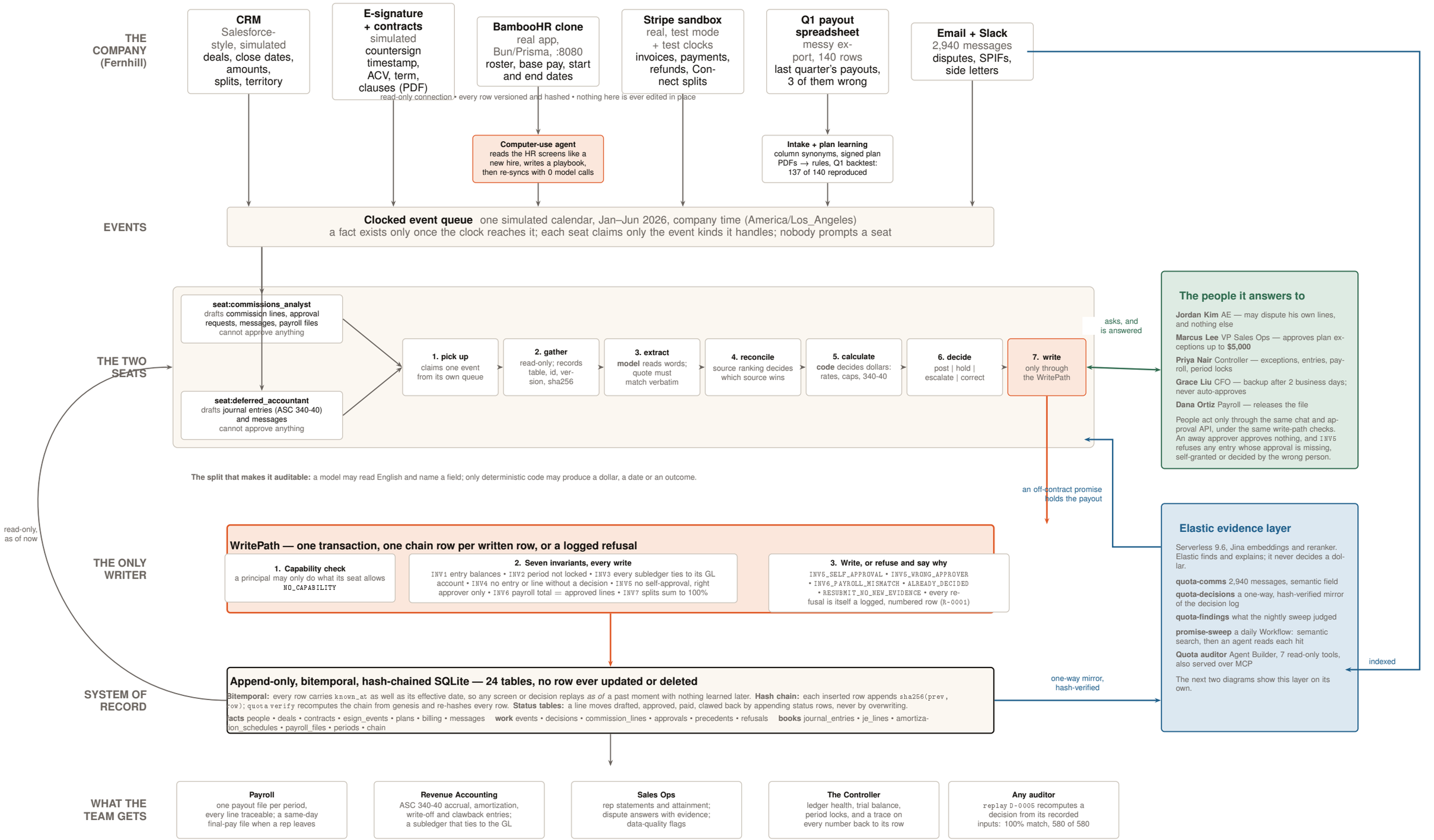

THE COMPANY (Fernhill)
CRM
Salesforce-style, simulated
deals, close dates, amounts, splits, territory
E-signature + contracts
simulated countersign timestamp, ACV, term, clauses (PDF)
BambooHR clone
real app, Bun/Prisma, :8080
roster, base pay, start and end dates
Stripe sandbox
real, test mode + test clocks
invoices, payments, refunds, Connect splits
Q1 payout spreadsheet
messy export, 140 rows
last quarter's payouts, 3 of them wrong
Email + Slack
2,940 messages
disputes, SPIFs, side letters
read-only connection • every row versioned and hashed • nothing here is ever edited in place
Computer-use agent
reads the HR screens like a new hire, writes a playbook, then re-syncs with 0 model calls
Intake + plan learning
column synonyms, signed plan PDFs → rules, Q1 backtest: 137 of 140 reproduced
EVENTS
Clocked event queue one simulated calendar, Jan–Jun 2026, company time (America/Los_Angeles)
a fact exists only once the clock reaches it; each seat claims only the event kinds it handles; nobody prompts a seat
THE TWO SEATS
seat:commissions_analyst
drafts commission lines, approval requests, messages, payroll files
cannot approve anything
seat:deferred_accountant
drafts journal entries (ASC 340-40) and messages
cannot approve anything
1. pick up
claims one event from its own queue
2. gather
read-only; records table, id, version, sha256
3. extract
model reads words; quote must match verbatim
4. reconcile
source ranking decides which source wins
5. calculate
code decides dollars: rates, caps, 340-40
6. decide
post | hold | escalate | correct
7. write
only through the WritePath
asks, and is answered
The people it answers to
Jordan Kim AE — may dispute his own lines, and nothing else
Marcus Lee VP Sales Ops — approves plan exceptions up to $5,000
Priya Nair Controller — exceptions, entries, payroll, period locks
Grace Liu CFO — backup after 2 business days; never auto-approves
Dana Ortiz Payroll — releases the file
People act only through the same chat and approval API, under the same write-path checks. An away approver approves nothing, and INV5 refuses any entry whose approval is missing, self-granted or decided by the wrong person.
The split that makes it auditable: a model may read English and name a field; only deterministic code may produce a dollar, a date or an outcome.
an off-contract promise holds the payout
read-only, as of now
THE ONLY WRITER
WritePath — one transaction, one chain row per written row, or a logged refusal
1. Capability check
a principal may only do what its seat allows
NO_CAPABILITY
2. Seven invariants, every write
INV1 entry balances • INV2 period not locked • INV3 every subledger ties to its GL account • INV4 no entry or line without a decision • INV5 no self-approval, right approver only • INV6 payroll total = approved lines • INV7 splits sum to 100%
3. Write, or refuse and say why
INV5_SELF_APPROVAL • INV5_WRONG_APPROVER • INV6_PAYROLL_MISMATCH • ALREADY_DECIDED • RESUBMIT_NO_NEW_EVIDENCE • every refusal is itself a logged, numbered row (R-0001)
Elastic evidence layer
Serverless 9.6, Jina embeddings and reranker. Elastic finds and explains; it never decides a dollar.
quota-comms 2,940 messages, semantic field
quota-decisions a one-way, hash-verified mirror of the decision log
quota-findings what the nightly sweep judged
promise-sweep a daily Workflow: semantic search, then an agent reads each hit
Quota auditor Agent Builder, 7 read-only tools, also served over MCP
The next two diagrams show this layer on its own.
indexed
SYSTEM OF RECORD
Append-only, bitemporal, hash-chained SQLite — 24 tables, no row ever updated or deleted
Bitemporal: every row carries known_at as well as its effective date, so any screen or decision replays as of a past moment with nothing learned later. Hash chain: each inserted row appends sha256(prev_row); quota verify recomputes the chain from genesis and re-hashes every row. Status tables: a line moves drafted, approved, paid, clawed back by appending status rows, never by overwriting.
facts people • deals • contracts • esign_events • plans • billing • messages work events • decisions • commission_lines • approvals • precedents • refusals books journal_entries • je_lines • amortization_schedules • payroll_files • periods • chain
one-way mirror, hash-verified
WHAT THE TEAM GETS
Payroll
one payout file per period, every line traceable; a same-day final-pay file when a rep leaves
Revenue Accounting
ASC 340-40 accrual, amortization, write-off and clawback entries; a subledger that ties to the GL
Sales Ops
rep statements and attainment; dispute answers with evidence; data-quality flags
The Controller
ledger health, trial balance, period locks, and a trace on every number back to its row
Any auditor
replay D-0005 recomputes a decision from its recorded inputs: 100% match, 580 of 580